Trajectory Optimization for a 3-DOF Robot Manipulator

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Abstract

Industrial robots repeat the same motions thousands of times per shift, so faster motions increase throughput. However, minimum-time motions are jerky, causing vibration, tracking error, and actuator wear. This project formulates a nonlinear program that plans the trajectory of a 3-DOF manipulator as a cubic spline through three intermediate waypoints, minimizing total motion time subject to joint-position, velocity, acceleration, jerk, and torque limits. We expect the minimum motion time to remain unchanged while the jerk limit is loose and other limits are tight, then to rise once the jerk limit becomes tight, with the motion time growing much more slowly than the jerk limit is reduced. We will quantify this speed–smoothness tradeoff by varying the jerk limit and validate the optimized trajectories in simulation.

1 Introduction

Minimizing motion time is a classic problem in manipulator trajectory planning. Purely time-optimal trajectories typically keep at least one actuator at its limit and switch abruptly between limits, producing discontinuous torques [1]. These abrupt changes can increase vibration in the links and drive train, and a feedback controller cannot follow them accurately. One approach to improving smoothness is to impose a hard upper bound on jerk [1, 2]. Stricter jerk limits, however, require longer motions. This project investigates the tradeoff between motion time and smoothness by solving a minimum-time trajectory optimization problem for a range of allowable jerk values.

We consider a rigid, non-planar 3-DOF manipulator whose base rotates about the vertical axis (joint 1), followed by a shoulder (joint 2) and an elbow (joint 3). The arm starts and ends at rest at fixed joint configurations. Its path is defined in joint space by three intermediate waypoints, with a cubic segment between each consecutive pair. The decisions are the nine waypoint joint angles (three joints at three waypoints) and the four segment durations, which together fix the entire motion.

Choosing these values is nontrivial for three reasons. First, the decisions are coupled. Because velocity and acceleration must be continuous, moving one waypoint or changing one duration changes the velocity at every waypoint and reshapes the whole motion. Second, the jerk limit conflicts directly with the objective. Halving every segment duration multiplies the jerk by eight, and which of the velocity, acceleration, jerk, and torque limits become tight, and where, is not known in advance. Third, actuator torques depend nonlinearly on the arm's configuration through gravity and inertia, so a straight line in joint space is not necessarily the fastest feasible path. Obstacle avoidance is excluded to keep the initial model tractable.

2 Problem Statement

2.1 Decision variables and trajectory

Let $i \in \{1, 2, 3\}$ index joints. Waypoints are numbered 0 (start) to 4 (goal), and segment $k \in \{1, 2, 3, 4\}$ runs from waypoint $k-1$ to waypoint k. The decision vector is

$$\mathbf{x} = [q_1^{(1)}, q_2^{(1)}, q_3^{(1)},\ q_1^{(2)}, q_2^{(2)}, q_3^{(2)},\ q_1^{(3)}, q_2^{(3)}, q_3^{(3)},\ h_1, h_2, h_3, h_4]^{\mathsf{T}} \in \mathbb{R}^{13},$$

where $q_i^{(k)}$ is the angle of joint i at intermediate waypoint k and h_k is the duration of segment k. The start and goal angles $q_i^{(0)}$ and $q_i^{(4)}$ are fixed, and the total time is $T = \sum_{k=1}^{4} h_k$. Using a clamped cubic spline [3], joint i on segment k, with local time $s \in [0, h_k]$, follows

$$q_{i,k}(s; \mathbf{x}) = q_i^{(k-1)} + v_i^{(k-1)} s + c_{i,k} s^2 + d_{i,k} s^3,$$

where $v_i^{(k)}$ is the joint velocity at waypoint k, with $v_i^{(0)} = v_i^{(4)} = 0$. Requiring continuous acceleration at the three intermediate waypoints gives three linear equations per joint, which are solved for the intermediate velocities at every evaluation. The coefficients $c_{i,k}(\mathbf{x})$ and $d_{i,k}(\mathbf{x})$ then follow in closed form (Appendix A). The resulting trajectory has continuous position, velocity, and acceleration, and its jerk is constant on each segment, $j_{i,k} = 6d_{i,k}(\mathbf{x})$.

2.2 Objective function

The objective is the total motion time,

$$\boxed{\min_{\mathbf{x}} \; f(\mathbf{x}) = T = \sum_{k=1}^{4} h_k.}$$

Jerk does not appear in the objective. Instead, a maximum allowable jerk is imposed as a hard constraint, and this limit is varied parametrically to measure how much motion time a given level of smoothness costs.

2.3 Constraints and model summary

All inequality constraints are written in negative null form. Each g_j is a family of scalar inequalities over joints $i = 1, 2, 3$, segments $k = 1, \ldots, 4$, and local times $s \in [0, h_k]$. At $s = 0$, the position limits g_1 and g_2 reduce to simple bounds on the waypoint angles, $q_{i,\min} \le q_i^{(k)} \le q_{i,\max}$, and g_3 and g_4 are simple bounds on the durations.

minimize	$f(\mathbf{x}) = \sum_{k=1}^{4} h_k$	total motion time
w.r.t.	$\mathbf{x} = \left[q_i^{(k)},\, h_k\right] \in \mathbb{R}^{13}$	waypoint angles, segment durations
subject to	$g_1(\mathbf{x}) = q_{i,\min} - q_{i,k}(s;\mathbf{x}) \le 0$	joint angle above lower limit
	$g_2(\mathbf{x}) = q_{i,k}(s;\mathbf{x}) - q_{i,\max} \le 0$	joint angle below upper limit
	$g_3(\mathbf{x}) = h_{\min} - h_k \le 0$	segment not too short
	$g_4(\mathbf{x}) = h_k - h_{\max} \le 0$	segment not too long
	$g_5(\mathbf{x}) = \dot{q}_{i,k}(s;\mathbf{x}) - v_{i,\max} \le 0$	velocity limit, positive direction
	$g_6(\mathbf{x}) = -\dot{q}_{i,k}(s;\mathbf{x}) - v_{i,\max} \le 0$	velocity limit, negative direction
	$g_7(\mathbf{x}) = \ddot{q}_{i,k}(s;\mathbf{x}) - a_{i,\max} \le 0$	acceleration limit, positive direction
	$g_8(\mathbf{x}) = -\ddot{q}_{i,k}(s;\mathbf{x}) - a_{i,\max} \le 0$	acceleration limit, negative direction
	$g_9(\mathbf{x}) = \tau_{i,k}(s;\mathbf{x}) - \tau_{i,\max} \le 0$	torque limit, positive direction
	$g_{10}(\mathbf{x}) = -\tau_{i,k}(s;\mathbf{x}) - \tau_{i,\max} \le 0$	torque limit, negative direction
	$g_{11}(\mathbf{x}) = 6d_{i,k}(\mathbf{x}) - j_{\max} \le 0$	jerk limit, positive direction
	$g_{12}(\mathbf{x}) = -6d_{i,k}(\mathbf{x}) - j_{\max} \le 0$	jerk limit, negative direction

where $q_{i,k}$ is the spline position from Section 2.1, $\dot{q}_{i,k}$, $\ddot{q}_{i,k}$, and $d_{i,k}$ are its velocity, acceleration, and cubic coefficient (Appendix A), and $\tau_{i,k}$ is joint i's torque from rigid-body inverse dynamics [4],

$$\boldsymbol{\tau}_k(s;\mathbf{x}) = \mathbf{M}(\mathbf{q}_k)\ddot{\mathbf{q}}_k + \mathbf{C}(\mathbf{q}_k,\dot{\mathbf{q}}_k)\dot{\mathbf{q}}_k + \boldsymbol{\tau}_{\mathrm{G}}(\mathbf{q}_k), \qquad \mathbf{q}_k(s) = [q_{1,k}(s), q_{2,k}(s), q_{3,k}(s)]^{\mathsf{T}}.$$

Here $\mathbf{M}$ is the mass matrix, $\mathbf{C}\dot{\mathbf{q}}$ contains the Coriolis and centrifugal terms, and $\boldsymbol{\tau}_{\mathrm{G}}$ is the gravity torque vector. There are no equality constraints. The acceleration-continuity equations are solved for the intermediate waypoint velocities and substituted into the model, and position and velocity continuity hold by construction.

For numerical solution, position, velocity, and torque limits are enforced at $N+1$ evenly spaced sample times per segment, $s = mh_k/N$ for $m = 0, \ldots, N$. Acceleration is linear in s, so its limits are enforced exactly at the waypoints. Jerk is constant on each segment, so its limits are enforced exactly once per segment. The final trajectory is rechecked on a finer grid.

3 Model Parameters

The robot will be the first three joints (base, shoulder, and elbow) of a commercial industrial arm, with the wrist joints held fixed. All parameter values will be taken from the selected robot's published specifications and models, as listed in Table 1. Parameters that manufacturers often do not publish, such as joint acceleration limits, will be estimated with stated assumptions (Section 4.6). The vector $\boldsymbol{\theta}_{\mathrm{arm}}$ contains the link lengths, masses, centers of mass, and inertias needed for inverse dynamics.

Table 1: Fixed parameters and their sources

Symbol	Description	Value or source	Units
$q_i^{(0)}$, $q_i^{(4)}$	Start and goal joint angles	Chosen within the robot's workspace	rad
$v_i^{(0)}$, $v_i^{(4)}$	Start and goal joint velocities	0 (start and end at rest)	rad/s
$q_{i,\min}$, $q_{i,\max}$	Joint-angle limits	Manufacturer specifications	rad
$v_{i,\max}$	Maximum joint speed	Manufacturer specifications	rad/s
$a_{i,\max}$	Maximum joint acceleration	Operating limit; estimated if unpublished	rad/s^2
$\tau_{i,\max}$	Maximum actuator torque	Manufacturer specifications, less friction margin	N m
$j_{\max}$	Maximum allowable joint jerk	Chosen limit; varied parametrically	rad/s^3
$h_{\min}$, $h_{\max}$	Segment-duration bounds	Set from initial feasibility tests	s
N	Sub-intervals per segment	10; increased until converged	–
$\boldsymbol{\theta}_{\mathrm{arm}}$	Robot geometry and inertias	Manufacturer's published dynamic model	mixed

4 Analysis of the Problem Statement

4.1 Problem size

The model has **13** continuous decision variables, **0** equality constraints, and 12 inequality constraint families. Each family expands to one scalar inequality per joint and per point where it is checked. With $N = 10$ there are $4N + 1 = 41$ sample times per joint, and the start and goal samples are dropped for position and velocity because their values are fixed. This gives **776** scalar inequalities: 234 for position, 8 for duration, 234 for velocity, 30 for acceleration, 246 for torque, and 24 for jerk (Appendix B).

4.2 Feasible point

Because the robot has not yet been selected, we demonstrate a feasible point using hypothetical parameter values (Table 8, Appendix C) that will not be used in the study. Place the waypoints evenly on the straight line from start to goal, $q_i^{(k)} = q_i^{(0)} + \frac{k}{4}\Delta q_i$ with $\Delta q_i = q_i^{(4)} - q_i^{(0)}$, and give all segments the same duration $h_k = T_0/4$. With start and goal angles $(0, 0, 0)$ and $(0.6, -0.4, 0.2)$ rad and $T_0 = 4$ s,

$$\mathbf{x}_0 = [0.15,\ -0.10,\ 0.05,\ \ 0.30,\ -0.20,\ 0.10,\ \ 0.45,\ -0.30,\ 0.15,\ \ 1,\ 1,\ 1,\ 1]^{\mathsf{T}}.$$

Table 2 shows that $\mathbf{x}_0$ satisfies every constraint. The peak velocity, acceleration, and jerk follow from closed-form expressions for this spline, and the torques are computed for a hypothetical arm (Appendix C). Slowing this motion reduces velocity, acceleration, jerk, and the inertial torques, while the gravity torque stays within the limits of any arm that can hold its own weight, so the same construction with a larger T_0 gives a feasible point for the selected robot.

Table 2: Constraint check at $\mathbf{x}_0$ (hypothetical parameters)

Constraints	Value at $\mathbf{x}_0$	Limit	Satisfied
g_1, g_2: joint position	between -0.4 and 0.6 rad	$[-1, 1]$ rad	Yes
g_3, g_4: segment duration	$h_k = 1$ s	$[0.5, 2]$ s	Yes
g_5, g_6: velocity	peak 0.206 rad/s	0.5 rad/s	Yes
g_7, g_8: acceleration	peak 0.514 rad/s^2	1.0 rad/s^2	Yes
g_9, g_{10}: torque	peaks 0.09, 4.65, 0.98 N m	8.0 N m	Yes
g_{11}, g_{12}: jerk	peak 0.643 rad/s^3	1.0 rad/s^3	Yes

4.3 Functions computed by separate routines

The intermediate velocities $v_i^{(k)}(\mathbf{x})$ come from a 3×3 tridiagonal linear system per joint, which is strictly diagonally dominant and therefore always solvable for $h_k > 0$. The torques $\tau_{i,k}(s; \mathbf{x})$ are computed with the recursive Newton–Euler algorithm [4] from the robot parameters $\boldsymbol{\theta}_{\text{arm}}$. Because the continuity system couples all waypoints, every velocity, coefficient, and torque depends on all 13 variables. Gradients will be computed analytically where practical, otherwise by finite differences, and checked against finite differences.

4.4 Natural and practical constraints

Natural constraints express physical laws or limits that must hold for the motion to be executable. Practical constraints are imposed operating or numerical limits that keep the search manageable [5]. Any practical constraint that is tight at the optimum will be relaxed to measure its cost.

Table 3: Natural and practical constraints

Constraint	Type	Reasoning
Torque (g_9, g_{10})	Natural	Set by the equations of motion; the motor saturates beyond $\tau_{i,\max}$.
Position (g_1, g_2)	Natural	Mechanical joint stops.
Velocity (g_5, g_6)	Natural	Physical maximum motor speed.
Velocity continuity	Natural	A joint cannot change velocity instantly (built into the spline).
Acceleration continuity	Practical	Smoothness choice of the cubic spline (built into the spline).
Acceleration (g_7, g_8)	Practical	Operating limit; the physical limit is enforced through torque.
Jerk (g_{11}, g_{12})	Practical	Chosen smoothness specification, not a physical law.
Duration (g_3, g_4)	Practical	$h_{\min} > 0$ keeps the formulas defined; $h_{\max}$ bounds the search.

4.5 Modeling assumptions and range of validity

The model is defined only for $h_k > 0$ and is used within the bounds $q_{i,\min} \le q_i^{(k)} \le q_{i,\max}$ and $h_{\min} \le h_k \le h_{\max}$.

Table 4: Modeling assumptions and range of validity

Assumption	Valid when / how checked
Rigid links	At speeds that do not excite structural vibration. The jerk limit and $h_{\min}$ keep the motion in this range.
Ideal actuators (no friction, motor dynamics, or torque-rate limits)	Torque changes slowly relative to the motor response. Torque limits include a margin for unmodeled friction.
Known parameters, no payload	For the unloaded arm. Payload and parameter uncertainty are studied in Section 4.6.
Four cubic segments, three waypoints	Optimal only within this family. Re-solving with more waypoints checks this.
No obstacles or self-collision	Each optimized trajectory is checked for collisions in simulation.
Limits hold between sample times	Each final trajectory is rechecked on a finer grid, and N is increased if needed.

4.6 Parameter determination, uncertainty, and parametric studies

Table 5 shows how each parameter will be determined and studied. In addition, the Lagrange multipliers of tight constraints give the local sensitivity of the minimum time to each limit, which will be checked by perturbing the parameters.

Table 5: How parameters are determined and studied

Parameter	How determined	Study
$j_{\max}$	Chosen smoothness specification	Sweep from loose to tight to trace minimum time versus jerk limit
$\tau_{i,\max}$ *(uncertain)*	Actuator specification, less a friction margin	Scale the limits and record the change in minimum time
Payload *(uncertain)*	Zero in the base model	Add an end-effector mass and increase it
$v_{i,\max}$, $a_{i,\max}$	Robot specification; selected operating limit	Multiplier sensitivity
$\boldsymbol{\theta}_{\text{arm}}$ *(uncertain)*	Published robot model or estimates	Perturb estimated parameters
Start and goal	Representative reach-and-place motion	Repeat for a second start–goal pair
$h_{\min}$, $h_{\max}$	Initial feasibility tests on the selected robot	Check whether either bound is tight
N; number of waypoints	Increased from initial values	Convergence checks on minimum time

4.7 Problem classification

This is a continuous **nonlinear program (NLP)** with 13 decision variables and no integer variables. The objective $T = \sum_k h_k$ is linear, so all of the nonconvexity comes from the constraints. The spline coefficients depend on inverse powers of the durations, and the torques depend on trigonometric functions of the joint angles, so the problem is **nonconvex** and may have multiple local minima. We will use

SLSQP [6] or IPOPT [7] from several initial points, including $\mathbf{x}_0$, and report the best feasible solution found, without a guarantee of global optimality.

4.8 Smoothness and numerical difficulties

Table 6: Smoothness and numerical difficulties

Question	Answer
Continuous and smooth?	Yes, for $h_k > 0$. All functions combine polynomials, inverse powers of h_k, a nonsingular linear solve, and smooth rigid-body dynamics. Writing each absolute-value limit as two inequalities keeps every constraint differentiable.
Undefined anywhere?	The spline coefficients are undefined at $h_k = 0$, which is excluded by $h_{\min} > 0$.
Gradients undefined?	No, within the duration bounds.
Regions without gradient information?	Yes. The objective depends only on the durations, so its gradient with respect to the nine waypoint angles is zero everywhere. The angles are determined only by the tight constraints, and if these do not fix them, many waypoint choices give the same minimum time, so the optimal angles are not unique. The minimum time T^* is still well defined. To report a single representative trajectory, we will re-solve with $T \le T^*$ while minimizing total squared jerk.
Poorly scaled?	Yes. For a fixed path shape, acceleration scales as h_k^{-2} and jerk as h_k^{-3}. Each constraint will be divided by its limit, for example $\tau_{i,k}/\tau_{i,\max} - 1 \le 0$.
Missed violations?	A position, velocity, or torque peak between sample times can be missed. Final trajectories are rechecked on a finer grid.

References

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A Spline construction

For each joint i, requiring continuous acceleration at intermediate waypoints $k = 1, 2, 3$ gives

$$\frac{v_i^{(k-1)}}{h_k} + 2\left(\frac{1}{h_k} + \frac{1}{h_{k+1}}\right) v_i^{(k)} + \frac{v_i^{(k+1)}}{h_{k+1}} = 3\left[\frac{q_i^{(k)} - q_i^{(k-1)}}{h_k^2} + \frac{q_i^{(k+1)} - q_i^{(k)}}{h_{k+1}^2}\right],$$

with $v_i^{(0)} = v_i^{(4)} = 0$. This is a 3×3 tridiagonal system in $v_i^{(1)}, v_i^{(2)}, v_i^{(3)}$. Each diagonal entry, $2(1/h_k + 1/h_{k+1})$, exceeds the sum of the off-diagonal entries in its row, so the matrix is strictly diagonally dominant and nonsingular for all $h_k > 0$. The coefficients of segment k are

$$c_{i,k} = \frac{3(q_i^{(k)} - q_i^{(k-1)})}{h_k^2} - \frac{2v_i^{(k-1)} + v_i^{(k)}}{h_k}, \qquad d_{i,k} = -\frac{2(q_i^{(k)} - q_i^{(k-1)})}{h_k^3} + \frac{v_i^{(k-1)} + v_i^{(k)}}{h_k^2},$$

which make each segment start and end at its waypoint angles with the shared waypoint velocities, so position and velocity are continuous by construction. The velocity, acceleration, and jerk are

$$\dot{q}_{i,k}(s) = v_i^{(k-1)} + 2c_{i,k}s + 3d_{i,k}s^2, \qquad \ddot{q}_{i,k}(s) = 2c_{i,k} + 6d_{i,k}s, \qquad j_{i,k} = 6d_{i,k}.$$

Acceleration is linear in s, so its extremes on each segment occur at the waypoints. Jerk is constant on each segment and may jump at the waypoints.

B Constraint count

With $N = 10$ sub-intervals per segment, each joint has $4N + 1 = 41$ sample times, since neighboring segments share their end times. The start and goal samples are dropped for position and velocity because those values are fixed parameters, leaving 39.

Table 7: Number of scalar inequality constraints ($N = 10$)

Constraint	Points checked	Count
g_1, g_2: joint position (lower, upper)	3 joints × 39 sample times, each	$117 + 117$
g_3, g_4: segment duration (lower, upper)	4 segments, each	$4 + 4$
g_5, g_6: velocity (positive, negative)	3 joints × 39 sample times, each	$117 + 117$
g_7, g_8: acceleration (positive, negative)	3 joints × 5 waypoints, each	$15 + 15$
g_9, g_{10}: torque (positive, negative)	3 joints × 41 sample times, each	$123 + 123$
g_{11}, g_{12}: jerk (positive, negative)	3 joints × 4 segments, each	$12 + 12$
Total		**776**

Of the position constraints, the 18 at the intermediate waypoints are the simple bounds on the waypoint angles. In general the total is $72N + 56$.

C Feasible point calculations

The values in Table 8 are hypothetical. They are used only to demonstrate a feasible point in Section 4.2 and will be replaced by the selected robot's parameters (Table 1).

Table 8: Hypothetical parameters for the feasible-point demonstration

Parameter	Value	Units
Start and goal angles $q_i^{(0)},\ q_i^{(4)}$	$(0,0,0)$ and $(0.6,-0.4,0.2)$	rad
Joint-angle limits $q_{i,\min},\ q_{i,\max}$	-1 and 1	rad
Velocity, acceleration, and jerk limits $v_{i,\max},\ a_{i,\max},\ j_{\max}$	0.5, 1.0, and 1.0	rad/s, rad/s^2, rad/s^3
Torque limit $\tau_{i,\max}$	8.0	N m
Segment-duration bounds $h_{\min},\ h_{\max}$	0.5 and 2.0	s

Kinematic limits. With evenly spaced waypoints, equal durations $h_k = T_0/4$, and zero end velocities, solving the system in Appendix A gives intermediate velocities of $\frac{9}{7}$, $\frac{6}{7}$, and $\frac{9}{7}$ times $\Delta q_i/T_0$. Every joint then follows the same curve scaled by Δq_i, which rises monotonically from start to goal without overshoot. Its peaks are

$$\max_t |\dot{q}_i| = \frac{48}{35}\frac{|\Delta q_i|}{T_0}, \qquad \max_t |\ddot{q}_i| = \frac{96}{7}\frac{|\Delta q_i|}{T_0^2}, \qquad \max_t |j_i| = \frac{480}{7}\frac{|\Delta q_i|}{T_0^3}.$$

The jerk peak occurs on the first and last segments. With $\max_i |\Delta q_i| = 0.6$ rad and $T_0 = 4$ s, these give 0.206 rad/s, 0.514 rad/s^2, and 0.643 rad/s^3, all below their limits. Because each joint is monotonic, it stays between its start and goal angles, which lie within $[-1, 1]$ rad, so the position limits hold. Each duration, $h_k = 1$ s, lies within $[0.5, 2]$ s.

Torque. The torques are computed from the equations of motion of a hypothetical arm. Joint 1 rotates the base about the vertical axis with rotational inertia 0.02 kg m^2. Joint 2 is the shoulder, with angle measured from horizontal, and carries a 0.25 m upper arm with a 1.0 kg point mass at the elbow. Joint 3 is the elbow, with angle measured relative to the upper arm, and carries a 0.20 m forearm with a 0.5 kg point mass at the tip. Evaluating the torques along $\mathbf{x}_0$ at 1000 times per segment gives peak magnitudes of 0.09, 4.65, and 0.98 N m for joints 1–3, all below $\tau_{i,\max} = 8$ N m. The shoulder torque is almost entirely gravity, because the accelerations along this slow trajectory are small. All values above were also confirmed numerically on the same fine grid. They will be recomputed once the robot is selected.